



Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

**This exam has 6 questions, worth 35 marks in total. You have 70 minutes.  
For multiple-choice questions, choose one answer.**

- 1) Consider a quadratic regression hypothesis: *[5 marks]*  
$$h\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$

Which feature appears in the gradient descent update for  $\theta_2$ ?

  - a) The first feature only.
  - b) The appropriate squared feature or validated estimate.
  - c) The number of parameters always decreases.
  - d) The test error is guaranteed to be zero.
- 2) Why can the matrix  $X^T X$  be non-invertible when fitting a linear regression model?  
*[5 marks]*
  - a) The first feature only.
  - b) The appropriate squared feature or validated estimate.
  - c) The number of parameters always decreases.
  - d) The test error is guaranteed to be zero.
- 3) A logistic regression model is trained with gradient descent. What can happen when the learning rate is too large? *[5 marks]*
  - a) The first feature only.
  - b) The appropriate squared feature or validated estimate.
  - c) The number of parameters always decreases.
  - d) The test error is guaranteed to be zero.
- 4) What happens as the degree of a polynomial model increases? Choose the statement that best describes overfitting. *[5 marks]*
  - a) The first feature only.
  - b) The appropriate squared feature or validated estimate.
  - c) The number of parameters always decreases.
  - d) The test error is guaranteed to be zero.

- 5) Which procedure gives the most reliable estimate of generalization performance when selecting a regularization strength? *[5 marks]*
- a) The first feature only.
  - b) The appropriate squared feature or validated estimate.
  - c) The number of parameters always decreases.
  - d) The test error is guaranteed to be zero.
- 6) Explain how you would detect overfitting and choose a regularization strength. Include the role of a validation set. *[10 marks]*